

Lecture 7: Prediction-Powered Inference

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We are covering *Prediction-Powered Inference* [Angelopoulos, Bates, Fannjiang, Jordan, and Zrnic, 2023]. The presentation in class was given by student Joe Kook.

1 Warmup: Mean Estimation

The presentation begins with the simplest target of estimating the mean, where we have:

- a small labeled, or gold-standard, sample:

$$(X_i, Y_i)_{i=1}^n$$

- a large unlabeled sample, with $N \gg n$:

$$(\tilde{X}_i)_{i=1}^N$$

- and a predictor f that can predict for both samples to output $f(X_i)$ and $f(\tilde{X}_i)$.

The central question is: *can we build a valid, shorter confidence interval using both samples?*

There are three strategies.

- **Imputation:** use $\tilde{\theta}^f = N^{-1} \sum_{i=1}^N f(\tilde{X}_i)$. This is cheap and low variance, but biased if predictions are systematically wrong.
- **Classical inference:** use $\hat{\theta}^{\text{class}} = n^{-1} \sum_{i=1}^n Y_i$. This is valid, but wastes the unlabeled sample and can be wide.
- **Prediction-powered inference (PPI):** use predictions for precision and labeled data to correct prediction bias.

For the mean, PPI takes the closed form

$$\hat{\theta}^{\text{PP}} = \underbrace{\frac{1}{N} \sum_{i=1}^N f(\tilde{X}_i)}_{\text{large imputed estimate}} - \underbrace{\frac{1}{n} \sum_{i=1}^n \{f(X_i) - Y_i\}}_{\text{estimated bias / rectifier}}.$$

The unlabeled predictions provide a high-throughput estimate, while the labeled data estimate how much the model prediction must be shifted to match true labels. If the predictor is accurate, $f(X_i) - Y_i$ has small variance, so the correction is stable.

The asymptotic confidence interval is

$$\hat{\theta}^{\text{PP}} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{\sigma}_{f-Y}^2}{n} + \frac{\hat{\sigma}_f^2}{N}}, \quad \hat{\theta}^{\text{class}} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{\sigma}_Y^2}{n}}$$

Since $N \gg n$, the PPI width is mainly driven by $\hat{\sigma}_{f-Y}^2/n$. Thus accurate predictions can make PPI tighter than classical inference while still correcting bias.

2 General Setup and Recipe

The warmup can be generalized from means to population parameters such as quantiles and regression coefficients. We observe

$$(X, Y) = \{(X_i, Y_i)\}_{i=1}^n \sim P, \quad \{\tilde{X}_i\}_{i=1}^N \sim P,$$

where the unlabeled outcomes $\tilde{Y}_i$ are unobserved. A black-box predictor f is trained on separate data and applied to both samples. The goal is a confidence set C_α^{PP} satisfying

$$\mathbb{P}(\theta^* \in C_\alpha^{\text{PP}}) \geq 1 - \alpha.$$

The important point is that PPI does *not* assume f is unbiased, calibrated, or even a consistent estimator of Y . It only needs labeled data to estimate the relevant prediction error.

Prediction-powered recipe

1. Start with the imputed estimator or estimating equation based on $f(\tilde{X})$.
2. Define a **rectifier** Δ that measures the bias caused by replacing Y with $f(X)$.
3. Build a confidence set for Δ using labeled data, then combine it with the imputed part to get a valid confidence set for θ^* .

The rectifier is problem-specific. For means, it is average prediction error. For quantiles, it is an error in the CDF. For regression, it is an error in the score or gradient equation. This is why the method is not simply “use a good model,” but rather “use labeled data to correct the model in the direction that matters for the estimand.”

3 Convex Estimation Problem

The main theory is stated for estimands that solve a convex optimization problem:

$$\theta^* = \arg \min_{\theta \in \mathbb{R}^p} \mathbb{E}[\ell_\theta(X_i, Y_i)],$$

where ℓ_θ is convex in θ . Under mild regularity conditions, convexity implies the first-order condition

$$\mathbb{E}[g_{\theta^*}(X_i, Y_i)] = 0,$$

where g_θ is a subgradient of ℓ_θ with respect to θ . This covers means, medians, quantiles, linear regression, logistic regression, and related estimands.

The prediction-based gradient is

$$g_\theta^f = \mathbb{E}[g_\theta(X_i, f(X_i))],$$

and the rectifier is defined as

$$\Delta_\theta = \mathbb{E}\{g_\theta(X_i, Y_i) - g_\theta(X_i, f(X_i))\}.$$

Thus the true gradient decomposes as

$$\mathbb{E}[g_\theta(X_i, Y_i)] = g_\theta^f + \Delta_\theta.$$

Interpretation: the unlabeled data estimate the gradient we would get if predictions were labels; the labeled data estimate the correction needed to turn that prediction-based gradient into the true gradient. The true parameter is plausible when the corrected gradient can be zero.

4 Prediction-Powered Confidence Set and Theorem

For each candidate θ , build two confidence sets:

$$\mathbb{P}(\Delta_\theta \in R_\delta(\theta)) \geq 1 - \delta, \quad \mathbb{P}(g_\theta^f \in T_{\alpha-\delta}(\theta)) \geq 1 - (\alpha - \delta).$$

Here $R_\delta(\theta)$ comes from labeled data and $T_{\alpha-\delta}(\theta)$ comes from unlabeled predictions. Define

$$C_\alpha^{\text{PP}} = \{\theta : 0 \in R_\delta(\theta) + T_{\alpha-\delta}(\theta)\},$$

where $A + B = \{a + b : a \in A, b \in B\}$ is the Minkowski sum. Equivalently, we keep all θ values for which zero is compatible with “prediction-based gradient plus rectifier.”

Theorem 1: Convex estimation

If the convex estimation problem is nondegenerate and the two component confidence sets have the stated coverage, then

$$\mathbb{P}(\theta^* \in C_\alpha^{\text{PP}}) \geq 1 - \alpha.$$

The validity statement is model-agnostic: the predictor can be any black box, because labeled data are used to correct its contribution.

Proof sketch. Let

$$E = \{\Delta_{\theta^*} \in R_\delta(\theta^*)\} \cap \{g_{\theta^*}^f \in T_{\alpha-\delta}(\theta^*)\}.$$

By a union bound, $\mathbb{P}(E) \geq 1 - \alpha$. On E ,

$$\mathbb{E}[g_{\theta^*}(X, Y)] = \Delta_{\theta^*} + g_{\theta^*}^f \in R_\delta(\theta^*) + T_{\alpha-\delta}(\theta^*).$$

Nondegeneracy gives $\mathbb{E}[g_{\theta^*}(X, Y)] = 0$, so $0 \in R_\delta(\theta^*) + T_{\alpha-\delta}(\theta^*)$. Therefore $\theta^* \in C_\alpha^{\text{PP}}$ on E .

5 Asymptotic Counterpart

The paper also presents an asymptotic version where $n, N \rightarrow \infty$ with $n/N \rightarrow \rho \in (0, 1)$. Instead of splitting the error budget between two separate confidence sets, one can directly plug in empirical estimates:

$$\widehat{\Delta}_\theta = \frac{1}{n} \sum_{i=1}^n \{g_\theta(X_i, Y_i) - g_\theta(X_i, f(X_i))\}, \quad \widehat{g}_\theta^f = \frac{1}{N} \sum_{i=1}^N g_\theta(\widetilde{X}_i, f(\widetilde{X}_i)).$$

A typical coordinate-wise confidence set keeps θ such that

$$|\widehat{\Delta}_{\theta,j} + \widehat{g}_{\theta,j}^f| \leq z_{1-\alpha/(2p)} \sqrt{\widehat{\sigma}_{\Delta,j}^2/n + \widehat{\sigma}_{g,j}^2/N} \quad \text{for all } j \in [p].$$

The interpretation is the same as before, but the construction is written as a corrected empirical estimating equation plus a normal half-width. Bonferroni correction handles multiple coordinates.

6 Examples: Mean and Linear Regression

For mean estimation, the loss is $\ell_\theta(y) = \frac{1}{2}(y - \theta)^2$, so $g_\theta(y) = \theta - y$ and

$$\Delta = \mathbb{E}[f(X) - Y].$$

This recovers the warmup estimator exactly.

For linear regression,

$$\theta^* = \arg \min_{\theta \in \mathbb{R}^d} \mathbb{E}[(Y_i - X_i^\top \theta)^2].$$

Because OLS is linear in the response, the slides give a closed form correction:

$$\widehat{\theta}^{\text{PP}} = \widetilde{X}^+ f(\widetilde{X}) - X^+ \{f(X) - Y\},$$

where A^+ denotes the pseudoinverse. The first term is OLS using predicted labels on the large unlabeled sample; the second term removes the regression bias observed on labeled data. For a single coordinate j^* ,

$$C_\alpha^{\text{PP}}(j^*) = \widehat{\theta}_{j^*}^{\text{PP}} \pm z_{1-\alpha/2} \sqrt{V_{j^*j^*}/n + \widetilde{V}_{j^*j^*}/N},$$

where V and $\widetilde{V}$ are sandwich variance estimates for the correction term and imputed term.

Future Research

- **Optimality of CI width.**

- **Done:** PPI++ (Angelopoulos, Duchi, Zrnic 2023) always improves on classical – safety guarantee.
- **Partial:** Xu, Wittern, Shojaie (2025) characterize the semiparametric efficiency bound; achieve it under stronger assumptions.

- **Open:** efficient and assumption-light estimators for general M-estimation (quantiles, GLMs).
- **Conformal $\times$ PPI.**
 - **Partial:** conformal PPI (Sillag et al. 2025).
- **Causal inference with black-box models.**
 - **Partial:** PPI for RCTs w/ digital twin controls (Poulet et al. 2025); PPI for transfer across experiments via conditional calibration (Cai et al. 2025).
 - **Open:** CATE / heterogeneous effects, IV / LATE, observational identification, sensitivity analysis.

References

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023. doi: 10.1126/science.adi6000. URL <https://arxiv.org/abs/2301.09633>.